SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Dharshavardhan SB	**Reg. No.:**	192420039
Date:		**Duration:**	60 minutes

Code of Conduct

I **Dharshavardhan 192420039** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Dharshavardhan

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Project: Smart Conference Paper Submission and Review System

1. Project Vision and Stakeholders

1.1 Project Vision

The vision of the Smart Conference Paper Submission and Review System is to develop a centralized web-based platform that simplifies and automates the complete academic conference paper management process, including author registration, paper submission, reviewer allocation, peer review, revision tracking, acceptance decisions, and final paper collection. The system aims to reduce administrative workload, improve reviewer coordination, maintain confidentiality, and ensure timely completion of conference activities.

1.2 Project Objectives

- Provide secure author and reviewer registration.
- Enable online research paper submission and tracking.
- Automate reviewer assignment while managing conflicts of interest.
- Support review report submission and decision-making.
- Track revised and final camera-ready papers.
- Send automated notifications about deadlines and decisions.
- Generate conference reports and analytics.

1.3 Key Stakeholders

Stakeholder	Role and Responsibility
Authors	Register, submit papers, revise manuscripts, and upload final versions.
Conference Chair	Manages the conference, deadlines, and final decisions.
Reviewers	Evaluate assigned papers and submit review reports.
Program Committee	Coordinates reviewers and supports decision-making.
System Administrator	Manages users, access permissions, and system configuration.
Organizing Committee	Monitors submissions, schedules, and conference statistics.

2. Create Epics and User Stories

Epics represent major functional areas of the project. Each epic contains smaller user stories that can be developed and tested independently.

Epic	ID	User Story
User Registration and Authentication	US01	As an author, I want to register and log in securely, so that I can access the conference submission portal.
User Registration and Authentication	US02	As an administrator, I want to manage user roles, so that authors, reviewers, and chairs receive appropriate access permissions.
Paper Submission Management	US03	As an author, I want to upload my research paper and enter its details, so that I can submit it for conference consideration.
Paper Submission Management	US04	As an author, I want to view my submission status, so that I can track the progress of my paper.
Reviewer Allocation and Conflict Management	US05	As a conference chair, I want to assign papers to suitable reviewers, so that every paper receives an appropriate evaluation.
Reviewer Allocation and Conflict Management	US06	As a reviewer, I want to declare conflicts of interest, so that I do not review papers where impartiality may be compromised.
Review and Decision Management	US07	As a reviewer, I want to submit ratings, comments, and recommendations, so that the conference committee can evaluate each paper.
Review and Decision Management	US08	As a conference chair, I want to record acceptance or rejection decisions, so that authors receive official outcomes.
Revision and Final Paper Management	US09	As an author, I want to upload a revised paper, so that I can address reviewer feedback.
Revision and Final Paper Management	US10	As an author, I want to submit the final camera-ready paper, so that my accepted work is ready for conference publication.
Notifications and Analytics	US11	As a user, I want to receive automated notifications, so that I am informed about deadlines, assignments, and decisions.
Notifications and Analytics	US12	As an organizing committee member, I want to view conference analytics, so that I can monitor submission counts, review progress, and acceptance statistics.

3. Define Acceptance Criteria

Acceptance criteria specify the conditions that must be satisfied before a user story is considered complete.

ID / User Story	Given–When–Then Acceptance Criteria
US01 – Registration	Given a new user, when valid registration details are submitted, then an account must be created and the user must be able to log in.
US02 – Role Management	Given an administrator, when a user role is assigned, then the system must restrict access according to that role.

US03 – Paper Submission	Given a registered author, when valid paper details and a supported file are uploaded before the deadline, then the system must save the submission and generate a unique paper ID.
US04 – Submission Status	Given a submitted paper, when the author opens the dashboard, then the latest submission status must be displayed.
US05 – Reviewer Assignment	Given an unassigned paper, when the chair assigns eligible reviewers, then the assignments must be recorded and visible to the selected reviewers.
US06 – Conflict Management	Given a reviewer assigned to a paper, when the reviewer declares a conflict, then the system must flag the conflict and prevent the conflicted reviewer from reviewing that paper.
US07 – Review Submission	Given an assigned reviewer, when all mandatory ratings, comments, and recommendations are submitted, then the review must be saved and linked to the correct paper.
US08 – Decision Management	Given completed reviews, when the authorized chair records a decision, then the decision must be saved and made available to the author.
US09 – Revision Submission	Given an author whose paper requires revision, when a valid revised version is uploaded before the revision deadline, then the system must store the new version while retaining the previous version.
US10 – Final Paper	Given an accepted paper, when the author uploads the final version before the deadline, then the system must record the camera-ready submission.
US11 – Notifications	Given a relevant event, when an assignment, deadline reminder, or decision is triggered, then the system must generate the corresponding notification for the intended user.
US12 – Analytics	Given conference submission and review records, when an authorized committee member opens the analytics dashboard, then the system must display accurate submission, review, and decision statistics.

4. Prioritize the Product Backlog

The backlog is prioritized using the MoSCoW technique: Must Have, Should Have, and Could Have. Dependencies and the importance of delivering a working submission workflow are also considered.

Priority	ID	User Story / Feature	Reason
P1 – Must Have	US01	Registration and login	Establishes secure access.
P1 – Must Have	US02	Role management	Protects restricted functions.
P1 – Must Have	US03	Paper submission	Core conference functionality.
P1 – Must Have	US04	Submission tracking	Provides status visibility.
P1 – Must Have	US05	Reviewer assignment	Enables the review workflow.
P1 – Must Have	US06	Conflict management	Supports impartial reviewing.
P1 – Must Have	US07	Review report submission	Enables paper evaluation.
P1 – Must Have	US08	Decision management	Communicates conference outcomes.
P2 – Should Have	US09	Revision submission	Supports author corrections.

P2 – Should Have	US10	Final paper collection	Prepares accepted papers for publication.
P2 – Should Have	US11	Automated notifications	Improves communication and deadline compliance.
P3 – Could Have	US12	Conference analytics	Supports reporting and planning.

Dependency order: Registration and roles → paper submission → reviewer assignment and conflict checks → review reports → decisions → revisions and final submission → analytics and reporting.

5. Plan the Sprint Backlog

Sprint 1 duration: 2 weeks (10 working days).

Sprint objective: Develop a functional foundation that enables users to register, log in, and submit research papers securely.

The first sprint focuses on essential features that can be implemented and tested within a short development cycle.

5.1 Selected User Stories

ID	User Story	Priority	Story Points
US01	User registration and login	Highest	5
US02	User role management	High	3
US03	Research paper submission	Highest	8
US04	Submission status tracking	High	3
TOTAL			19

5.2 Actionable Development Tasks

User Story	Development Tasks	Estimated Effort
US01	Design registration and login screens; implement account creation and password hashing; validate inputs; test authentication.	2 days
US02	Define user roles; implement role-based access control; test access restrictions.	1.5 days
US03	Design the submission form; implement paper upload and validation; save paper metadata and generate a unique paper ID; test upload handling.	4 days
US04	Design the author dashboard; retrieve submission records; display paper IDs and statuses; test status updates.	1.5 days

Integration	Integration testing, bug fixing, and sprint review preparation.	1 day
TOTAL		10 working days

The task durations are initial estimates. The team can revise them after reviewing technical dependencies and actual progress.

6. Estimate Effort Using Story Points

The Fibonacci sequence (1, 2, 3, 5, 8, 13, 21...) is used to estimate relative complexity, development effort, testing requirements, and uncertainty. The team can assign points using Planning Poker, where each member independently estimates a story and discusses differences before agreeing on a value.

6.1 Story Point Estimation

User Story	Story Points	Justification
US01 – Registration and login	5	Requires authentication, input validation, and password security.
US02 – Role management	3	Involves defining roles and enforcing access permissions.
US03 – Paper submission	8	Includes file validation, metadata storage, upload handling, and unique ID generation.
US04 – Status tracking	3	Requires retrieving submission records and displaying their current status.
TOTAL	19	Moderate initial sprint workload.

6.2 Estimation Technique

Planning Poker is selected because it encourages team discussion and reduces individual estimation bias.

- 3 points: Simple functionality with limited complexity.
- 5 points: Moderate functionality involving multiple components.
- 8 points: More complex functionality involving file handling, validation, and database integration.

Story points measure relative complexity rather than fixed hours. The team should confirm its capacity based on previous sprint performance; as this is the first sprint, 19 points is a provisional target.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

To develop the initial working version of the Smart Conference Paper Submission and Review System, enabling users to register and log in securely, enforcing role-based access, allowing authors to submit research papers, and providing submission status tracking.

7.2 Expected Deliverables

Deliverable	Expected Output
User Authentication Module	Registration, login, input validation, and secure password storage.
Role Management Module	Separate access permissions for authors, reviewers, conference chairs, and administrators.
Paper Submission Module	Submission form, file validation, metadata storage, and unique paper ID generation.
Author Dashboard	Displays submitted papers, unique IDs, and current submission statuses.
Tested Sprint Increment	Integrated, tested, and demonstrable software covering all four selected user stories, along with test results and a sprint review demonstration.

7.3 Definition of Done

- All agreed acceptance criteria are satisfied.
- Code is implemented and integrated successfully.
- Functional and relevant security tests pass.
- No critical defects remain in the completed functionality.
- The feature is demonstrated during the sprint review.
- Basic technical documentation is updated.

8. Conclusion

The Agile Sprint Planning and User Story Workshop establishes a structured development plan for the Smart Conference Paper Submission and Review System. By dividing the project into epics, user stories, acceptance criteria, and prioritized backlog items, the team can develop the system incrementally.

The first sprint focuses on authentication, role management, paper submission, and status tracking, with a provisional estimate of 19 story points over two weeks. Subsequent sprints can implement reviewer allocation, conflict-of-interest management, review reports, decision notifications, revisions, final paper collection, and conference analytics. This approach supports continuous testing, stakeholder feedback, and reliable delivery of the complete system.

9. REFERENCE

1. Stergiopoulos, V., Vassilakopoulos, M., Tousidou, E., Kavvathas, S., and Corral, A. (2025). Conference Management System Utilizing an LLM-Based Recommendation System for the Reviewer Assignment Problem. Proceedings of the International Conference on Enterprise Information Systems (ICEIS), 1004–1011.

2. Midde, R. R., Lingala, G. S., Dubasi, H., Mangala, G., and Pullaiahgari, P. K. R. (2025). AI-Enabled Academic Conference Management System. Proceedings of the International Conference on Research and Development in Information, Communication, and Computing Technologies, 328–334.

3. Zuccon, G. (2025). ReviewHQ: An API-Based System for Reviewer Assignment and Quality Control in Research Conferences. Proceedings of SIGIR 2025.

4. Bouanane, K., Medakene, A. N., Benbelghit, A., and Belhaouari, S. B. (2024). FairColor: An Efficient Algorithm for the Balanced and Fair Reviewer Assignment Problem. Information Processing & Management, 61(6), 103865.

5. Fu, Y., Luo, J., Nan, G., and Li, D. (2025). Peer Review Expert Group Recommendation: A Multi-Subject Coverage-Based Approach. Expert Systems with Applications, 264, 125971.

6. Leyton-Brown, K., Mausam, Nandwani, Y., Zarkoob, H., Cameron, C., Newman, N., and Raghu, D. (2024). Matching Papers and Reviewers at Large Conferences. Artificial Intelligence, 331, 104119.

7. Goldberg, A., Stelmakh, I., Cho, K., Oh, A., Agarwal, A., Belgrave, D., et al. (2025). Peer Reviews of Peer Reviews: A Randomized Controlled Trial and Other Experiments. PLOS ONE, 20(4), e0320444.

8. Idahl, M., and Ahmadi, Z. (2025). OpenReviewer: A Specialized Large Language Model for Generating Critical Scientific Paper Reviews. Proceedings of NAACL 2025: Demonstrations.

9. Leveraging Knowledge Graphs and LLMs for Content-Based Reviewer Assignment. (2025). Journal of Intelligent Information Systems.

10. Leveraging Large Language Models for Academic Conference Organization. (2025). npj Digital Medicine.